

File Structure

```
.
|-- README.md
|-- CSC220_File_Operations_TODO.txt
|-- src
|   |-- FileDemo1.java
|   |-- FileDemo2.java
|   |-- FileDemo3.java
|   |-- FileDemo4.java
|   |-- FileDemo5.java
|-- students.txt
`-- students2.txt
```

Compile

```
javac -d bin src/FileDemo3.java
```

Run

```
java -cp bin FileDemo3
```

Zip/compress the files

```
zip -r FileExamples.zip . -x "bin/*" "FileExamples.zip"
```

Notes:

PrintWriter: Overwrite vs. Append

PrintWriter has multiple **overloaded constructors**.

Java selects the appropriate constructor based on the type of argument we provide.

	Overwrite	Append
Code	<code>new PrintWriter(filename)</code>	<code>new PrintWriter(new FileWriter(filename, true))</code>
Argument passed to PrintWriter	String	FileWriter object

	Overwrite	Append
PrintWriter constructor used	<code>PrintWriter(String fileName)</code>	<code>PrintWriter(Writer out)</code>
If the file exists	Existing content is overwritten	New content is added to the end
Main checked exception	<code>FileNotFoundException</code>	<code>IOException</code>
Use when	Creating/replacing a file	Adding data without deleting existing data

Why are different constructors called?

Consider:

```
PrintWriter output = new PrintWriter(filename);
```

Here, `filename` is a `String`. Therefore, Java selects a `PrintWriter` constructor that accepts a `String`.

```
PrintWriter(String fileName)
```

Now consider:

```
PrintWriter output =
    new PrintWriter(new FileWriter(filename, true));
```

First, Java creates a `FileWriter`:

```
FileWriter fw = new FileWriter(filename, true);
```

Then that `FileWriter` object is passed to `PrintWriter`:

```
PrintWriter output = new PrintWriter(fw);
```

Since `FileWriter` is a type of `Writer`, Java selects the `PrintWriter` constructor that accepts a `Writer`.

FileWriter's Append Option

`FileWriter` itself has an overloaded constructor:

```
new FileWriter(filename, true)
```

The second argument controls whether data is appended.

Value	Behavior
<code>true</code>	Append new data to the existing file
<code>false</code>	Overwrite the existing file

Therefore:

```
new FileWriter(filename, true)
```

means:

Open the file for writing, but preserve the existing content and add new content at the end.

Key Idea

These two statements both create a `PrintWriter`, but they use different overloaded constructors:

```
new PrintWriter(filename)
      ↑
    String

new PrintWriter(new FileWriter(filename, true))
      ↑
    Writer object
```

Constructor overloading: Same constructor name, but different parameter types or parameter lists determine which constructor Java calls.

Another note:

- `PrintWriter` handles `FileNotFoundException` where as the overloaded version with `FileWriter` handles `IOException` because that's how they are defined in the Java API.
- Remember:

```
Exception
  ↑
IOException
  ↑
FileNotFoundException
```

So `FileNotFoundException` **IS-A** `IOException`.

- Which checked exception we must handle or declare is determined by the method or constructor we are calling. Look at its API documentation and check its throws declaration.
- Check: <https://docs.oracle.com/javase/8/docs/api/java/io/PrintWriter.html>

=> **`FileNotFoundException`** is the main/most common checked exception we see. But conceptually, file writing can fail because of things like:

- invalid or inaccessible path
- no write permission
- disk/device problem
- file system failure

=> Many lower-level writing APIs express these problems through IOException.